

Word Problems & Applications

Marking Scheme & Blueprint

MTH1W · Module 4 Test — Teacher Copy

Blueprint

Curriculum: MTH1W (Grade 9 de-streamed math). Total: 70 marks · 60 minutes · 30 questions. Categories: K/U, Application (App), Thinking (Think), Communication (Comm).

Q	Sub-topic	Level	Marks	Category	Section
A1	Translating	L1	1	K/U	A
A2	Translating	L1	1	K/U	A
A3	Translating	L1	1	K/U	A
A4	Percent	L1	1	K/U	A
A5	Distance-Rate-Time	L1	1	K/U	A
A6	Number	L1	1	K/U	A
A7	Geometry	L1	1	K/U	A
A8	Translating	L1	1	K/U	A
A9	Money	L1	1	K/U	A
A10	Number	L1	2	K/U	A
A11	Translating	L1	2	K/U	A
A12	Percent	L1	2	K/U	A
A13	Distance-Rate-Time	L1	2	K/U	A
A14	Geometry	L1	2	K/U	A
A15	Number	L1	2	K/U	A
B16	Translating	L2	2	App	B
B17	Number	L2	2	App	B
B18	Money	L2	2	App	B
B19	Mixture	L2	2	App	B
B20	Percent	L2	2	App	B
B21	Distance-Rate-Time	L2	2	App	B
B22	Number	L2	3	App	B
B23	Percent	L2	3	App	B
B24	Geometry	L2	3	App	B
C25	Money	L3	4	Think	C
C26	Number	L3	4	Think	C
C27	Mixture	L3	5	Think	C
C28	Geometry	L3	5	Think	C
C29	Distance-Rate-Time	L3	6	Think	C
D30	Percent	L3	4	Comm	D

By category (marks): K/U 21 · Application 21 · Thinking 24 · Communication 4

By level (marks): L1 21 · L2 21 · L3 28 → L1 30% · L2 30% · L3 40%

Worked Solutions & Mark Allocation

- 1. (A1 · K/U · 1 marks)**
 $x + 9$. (A1)
- 2. (A2 · K/U · 1 marks)**
 $6n$. (A1)
- 3. (A3 · K/U · 1 marks)**
 $n - 5 = 12$. (A1)
- 4. (A4 · K/U · 1 marks)**
42. (A1)
- 5. (A5 · K/U · 1 marks)**
40 km/h. (A1)
- 6. (A6 · K/U · 1 marks)**
 $n + 1$. (A1)
- 7. (A7 · K/U · 1 marks)**
 $4s$. (A1)
- 8. (A8 · K/U · 1 marks)**
 $2x + 3$. (A1)
- 9. (A9 · K/U · 1 marks)**
5c. (A1)
- 10. (A10 · K/U · 2 marks)**
 $n + 7 = 20$; $n = 13$. (M1, A1)
- 11. (A11 · K/U · 2 marks)**
 $3n = 27$; $n = 9$. (M1, A1)
- 12. (A12 · K/U · 2 marks)**
 $0.25 \times 48 = \$12$. (M1, A1)
- 13. (A13 · K/U · 2 marks)**
 $30 \div 3 = 10$ km/h. (M1, A1)
- 14. (A14 · K/U · 2 marks)**
 $2(6 + 4) = 20$ cm. (M1, A1)
- 15. (A15 · K/U · 2 marks)**
 $2n + 4 = 16 \rightarrow 2n = 12 \rightarrow n = 6$. (M1, A1)
- 16. (B16 · Application · 2 marks)**
 $4n - 7 = 29 \rightarrow 4n = 36 \rightarrow n = 9$. (M1, A1)

17. (B17 • Application • 2 marks)

$2n + 1 = 45 \rightarrow n = 22$; integers 22 and 23. (M1, A1)

18. (B18 • Application • 2 marks)

$4 + 2k = 20 \rightarrow 2k = 16 \rightarrow k = 8$ km. (M1, A1)

19. (B19 • Application • 2 marks)

$0.15 \times 400 = 60$ mL. (M1, A1)

20. (B20 • Application • 2 marks)

$90 - 0.30(90) = 90 - 27 = \63 . (M1, A1)

21. (B21 • Application • 2 marks)

$t = 96 \div 24 = 4$ hours. (M1, A1)

22. (B22 • Application • 3 marks)

$3n + 3 = 72 \rightarrow 3n = 69 \rightarrow n = 23$; integers 23, 24, 25. (M1, M1, A1)

23. (B23 • Application • 3 marks)

$1.13p = 79.10 \rightarrow p = \70 . (M1, M1, A1)

24. (B24 • Application • 3 marks)

$x + 4x = 90 \rightarrow 5x = 90 \rightarrow x = 18$; angles 18° and 72° . (M1, M1, A1)

25. (C25 • Thinking • 4 marks)

$30 + 0.05m = 47.50$. (M1)

$0.05m = 17.50$. (M1)

$m = 350$ minutes. (A1, A1)

26. (C26 • Thinking • 4 marks)

$n + (n+2) + (n+4) = 63$. (M1)

$3n + 6 = 63 \rightarrow 3n = 57 \rightarrow n = 19$. (M1, A1)

Integers 19, 21, 23. (A1)

27. (C27 • Thinking • 5 marks)

Let $x =$ kg of the \$7 coffee, $(15 - x) =$ kg of the \$13 coffee. (M1)

$7x + 13(15 - x) = 9(15) = 135$. (M1)

$7x + 195 - 13x = 135 \rightarrow -6x = -60 \rightarrow x = 10$. (A1, A1)

10 kg of \$7 coffee and 5 kg of \$13 coffee. (A1)

28. (C28 • Thinking • 5 marks)

Let $w =$ width, length $= 2w + 3$. $2(w + 2w + 3) = 60$. (M1)

$6w + 6 = 60 \rightarrow 6w = 54 \rightarrow w = 9$; length $= 21$. (M1, A1)

Dimensions 9 m by 21 m. (A1)

Area $= 9 \times 21 = 189 \text{ m}^2$. (A1)

29. (C29 • Thinking • 6 marks)

Combined closing rate = $18 + 22 = 40$ km/h. (M1)

$40t = 120 \rightarrow t = 3$ hours. (M1, A1)

First cyclist: $18 \times 3 = 54$ km. (A1, M1)

Second cyclist: $22 \times 3 = 66$ km. (A1) (Check: $54 + 66 = 120$.)

30. (D30 • Communication • 4 marks)

30% off \$100 gives \$70. (A1)

An extra 20% off \$70 gives $0.80 \times 70 = \$56$. (A1)

The discounts multiply ($0.70 \times 0.80 = 0.56$), they do not add, so the price is \$56. (C1)

\$56 paid means 44% off, not 50% off. (C1)